StarMap Workshop

Ahmed Eldawy, Associate Professor, UCR
Ibrahim Sabek, Assistant Professor, USC
Tarlan Bahadori, PhD Student, UCR
Shaolin (Shelly) Xie, PhD Student, USC

Welcome

- Goal: Teaching and giving hands-on experience on interactive exploration and visualization of large geospatial datasets
  - Learn how to work with geospatial data
  - Build visualizations
  - Experiment with styling workflows, including LLM-assisted approaches

- Educational and Fun Experience
  - Competition and winning prizes ☺

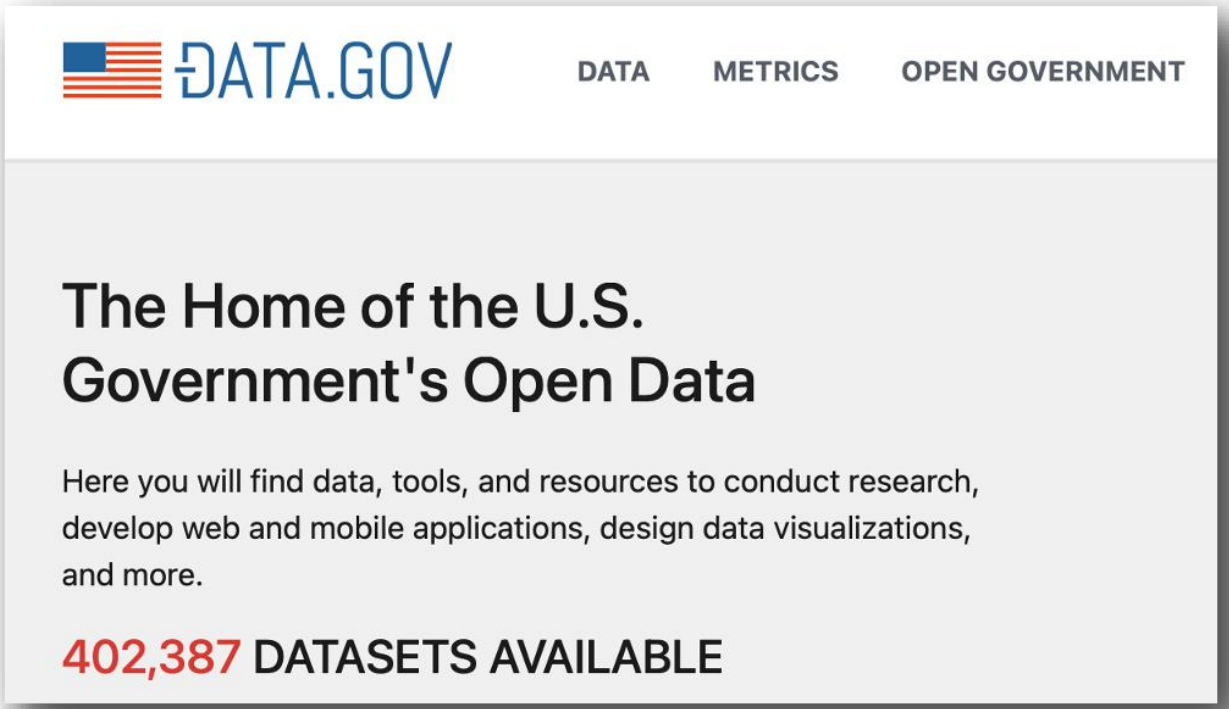
Introduction to GIS*

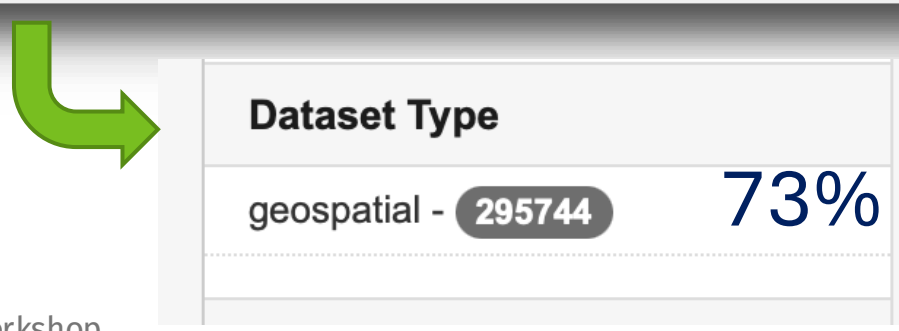
April 13, 2026

*Based on slides by Janet Reyes, Geospatial Information Librarian, UCR

Rise of Big Open Data

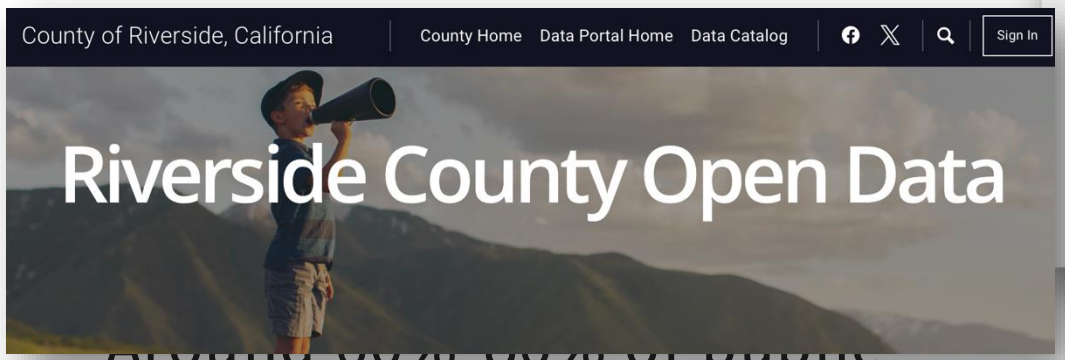
- Data.gov launched in May 2009
- OPEN Government Data Act unanimously approved by Senate in 2016
- All government agencies are required to publish their data in a readable format on Data.gov
- Around 60%-80% of public data is geospatial

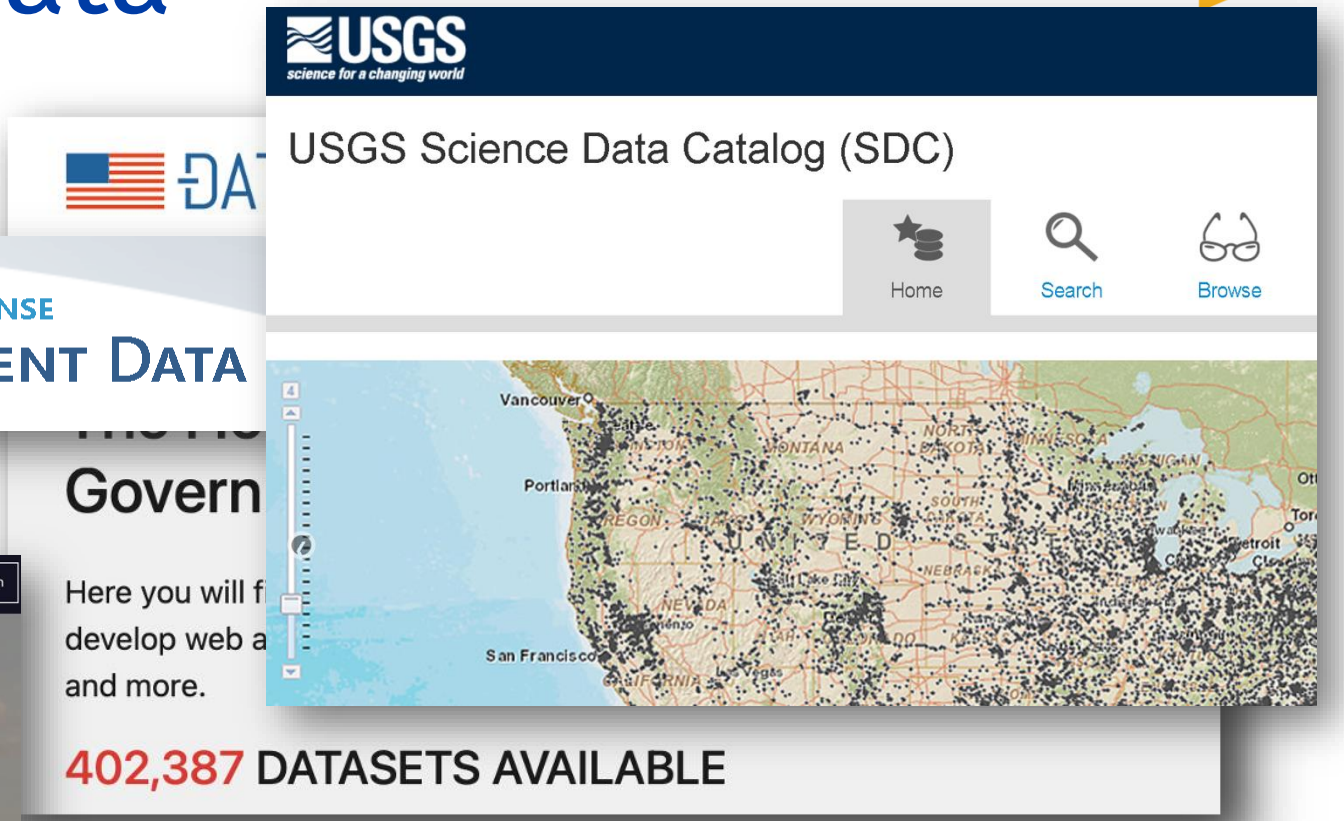




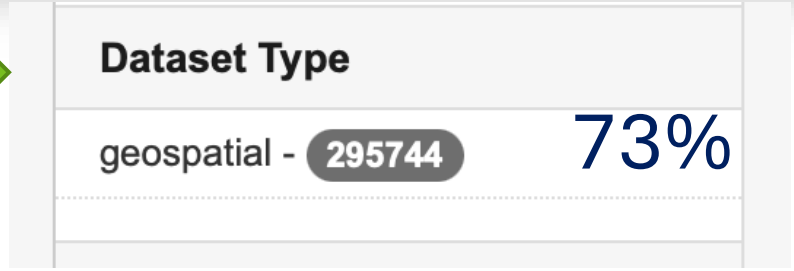
Rise of Big Open Data

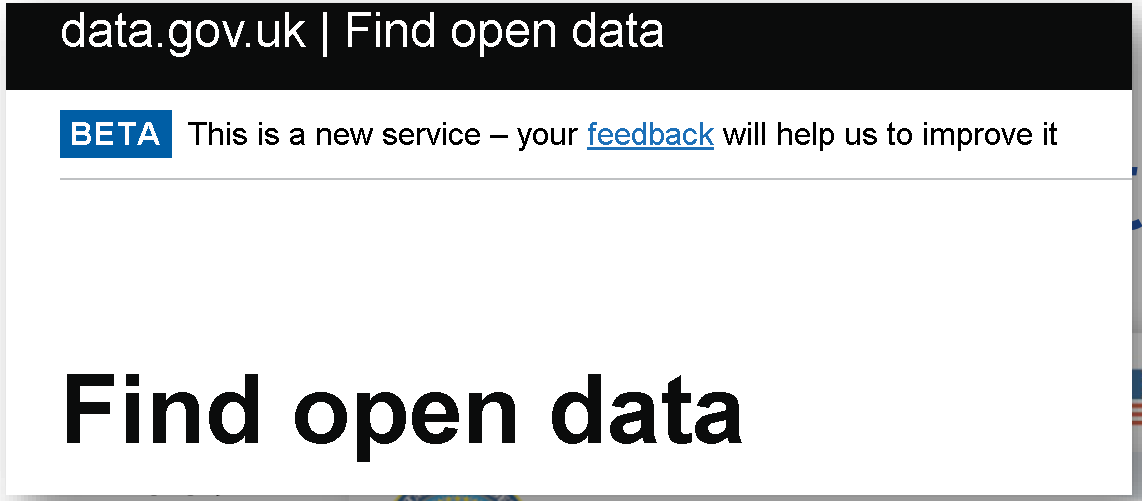
- Data.gov launched in May 2009
- OPEN Go
  unanimously approved by Senate in 2016





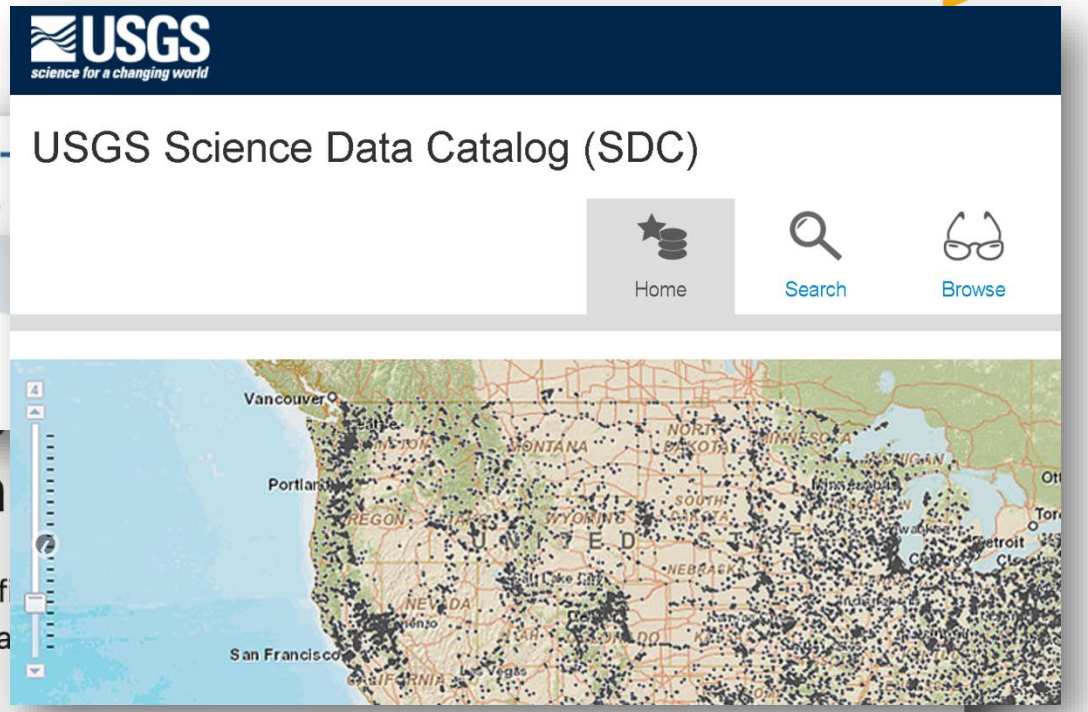
- Around 60% - 80% of public data is geospatial

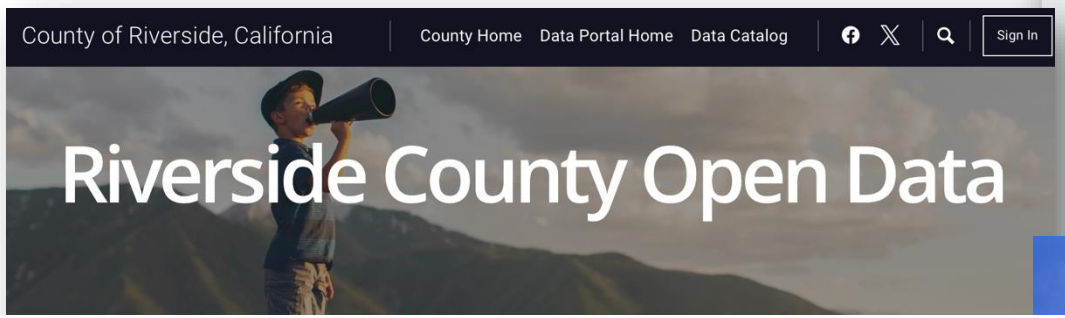




- OPEN Go
  unanimously approved by
  Senate in 2016

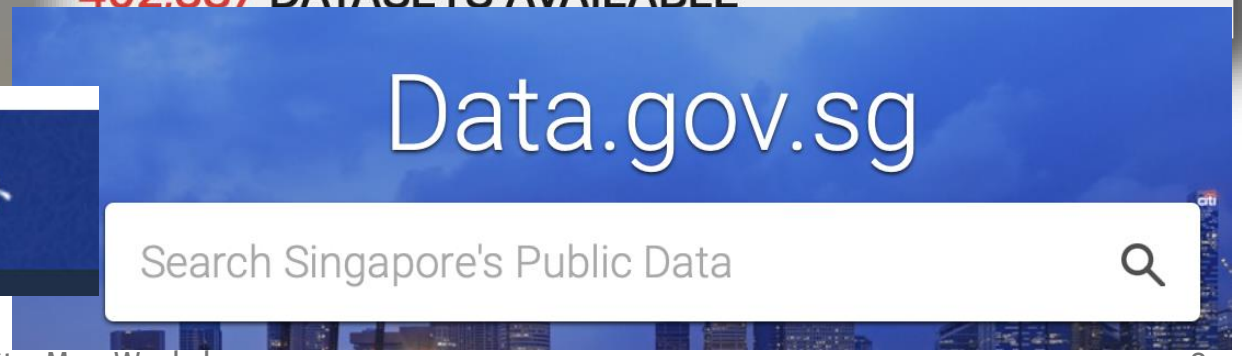
U.S. DEPARTMENT OF DEFENSE
OPEN GOVERNMENT DATA





Here you will f
develop web a
and more.

402,387 DATASETS AVAILABLE



Find open data

- OPEN
  unanir
  Senate

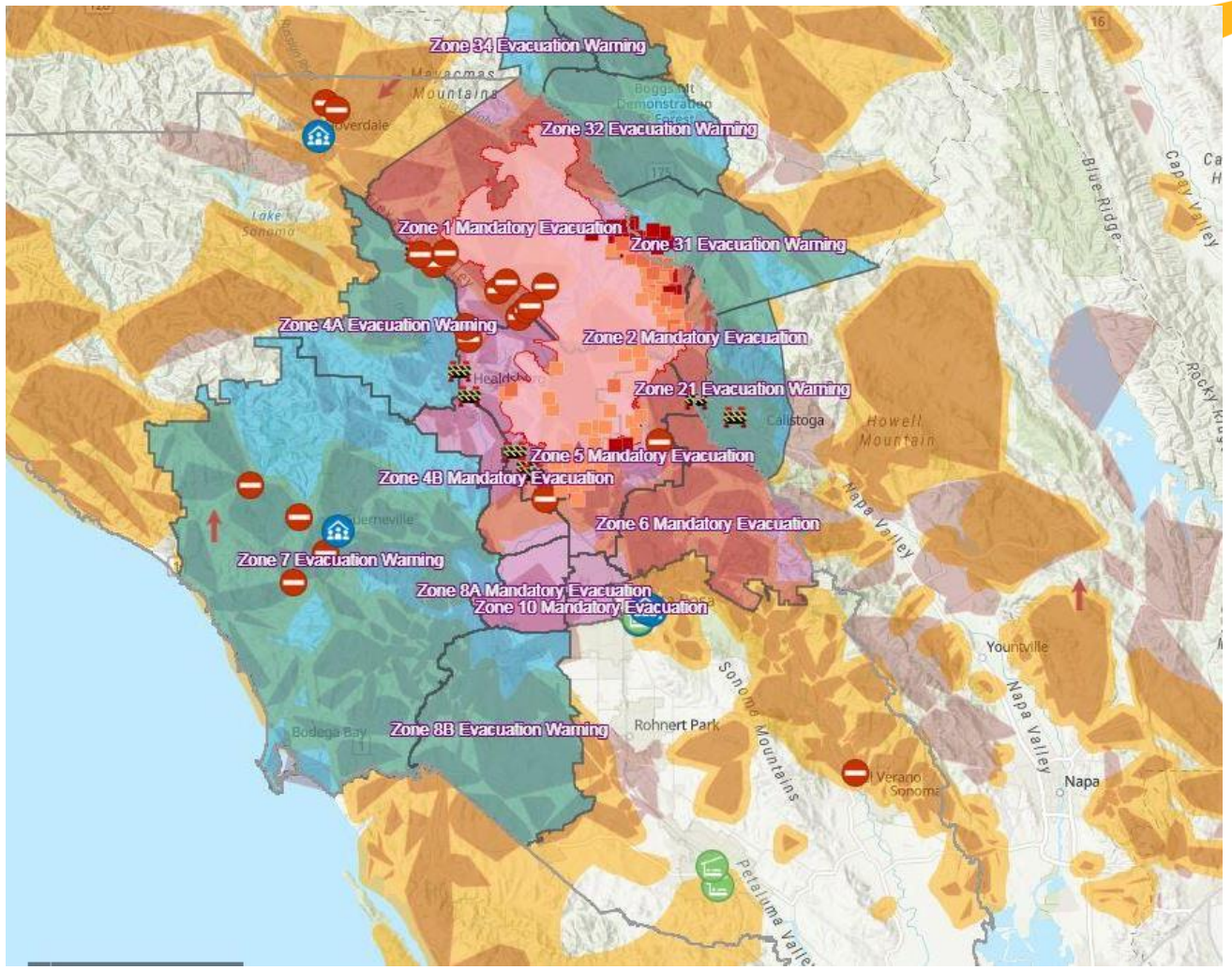
GIS for Wildfire Monitoring & Response

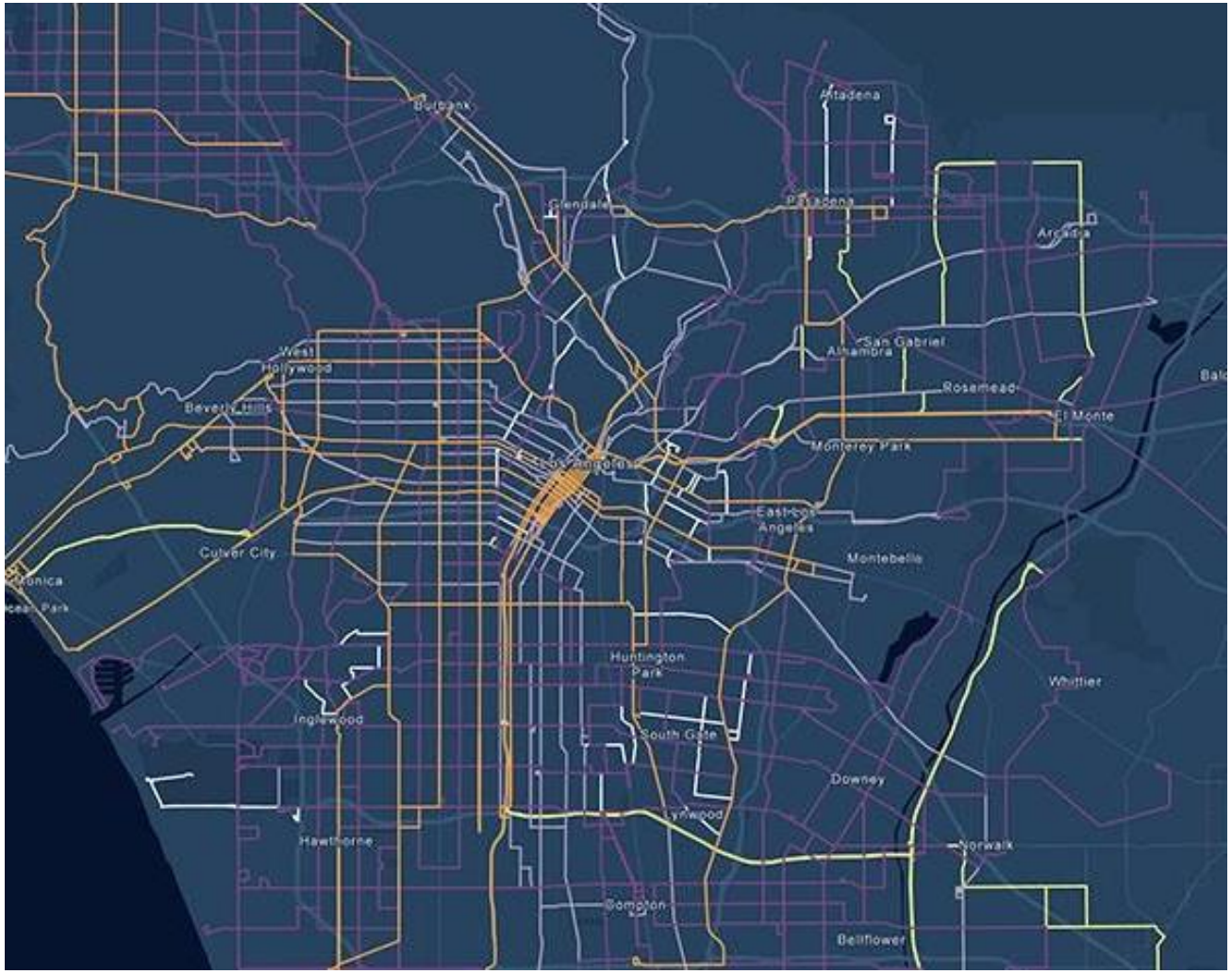
Maps active fires, evacuation zones, and risk areas in real time

Combines satellite data, weather, and terrain to track fire spread

Helps emergency responders coordinate resources and evacuations

Public tools (e.g., apps like Watch Duty) provide live GIS-based alerts





GIS for Traffic & Transportation

Critical for planning transportation, e.g., during the **2028 LA Olympics**

Predicts **congestion** and optimizes routes for millions of visitors

Coordinates transit, road closures, and event logistics

Uses real-time data to manage traffic and crowd movement

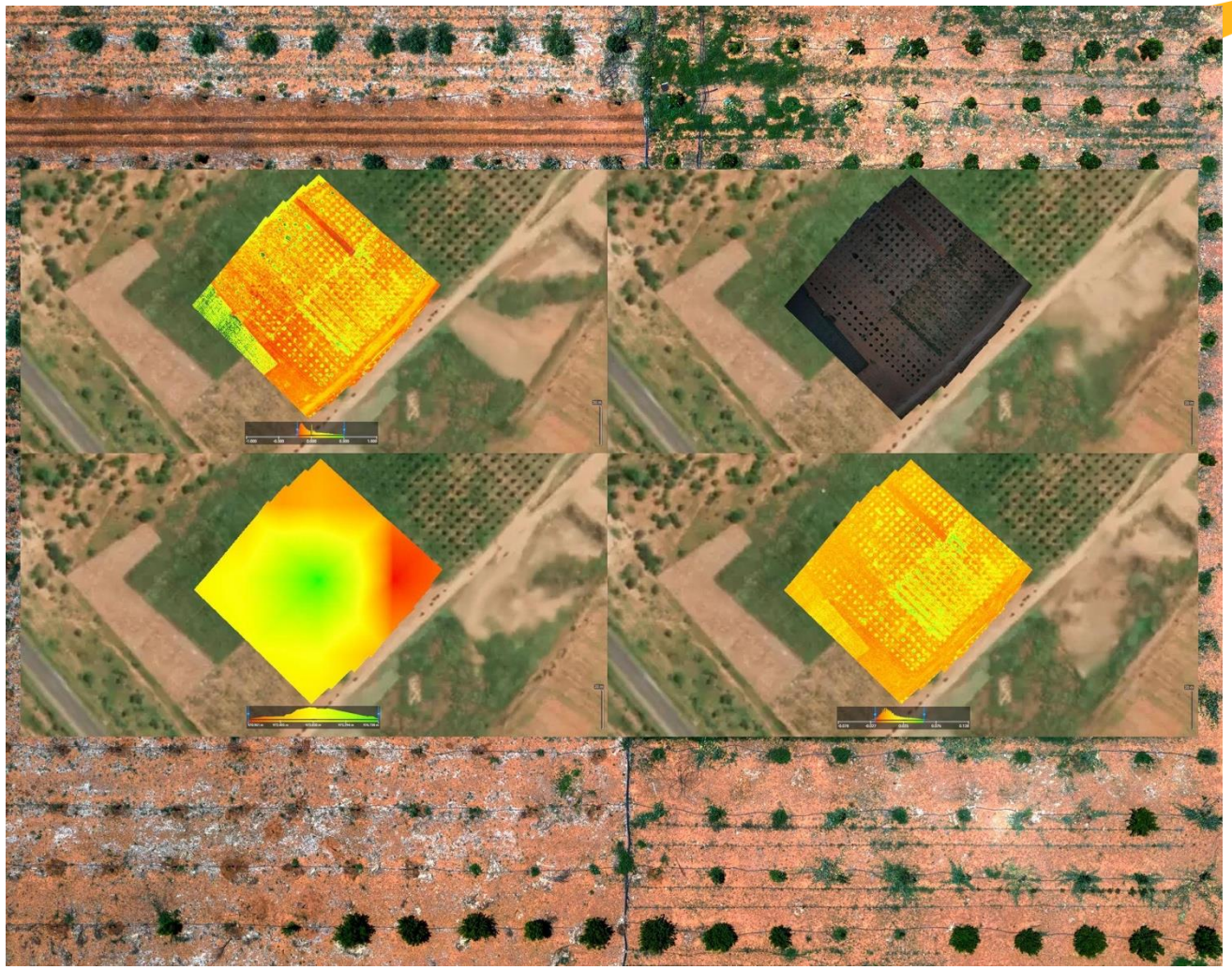
GIS for Agriculture & Food Production

Uses satellite and drone data to monitor crop health and growth

Identifies areas that need more or less water, fertilizer, or treatment

Helps farmers increase yield while reducing costs and environmental impact

Supports water management—critical in drought-prone California



Outline

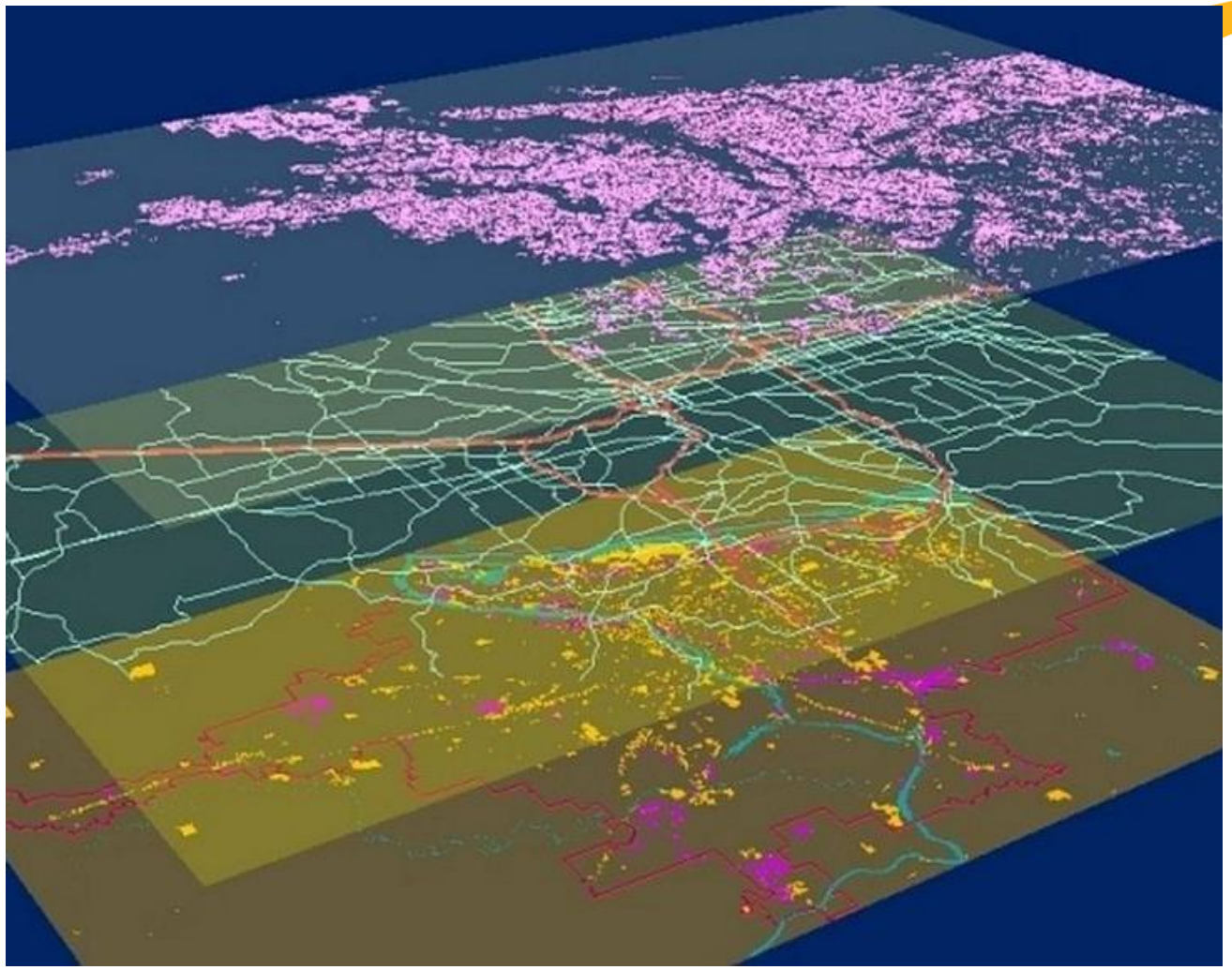
- Definitions
- What is GIS used for?
- Types of data used in GIS
- GIS software libraries
- Projections and coordinate systems
- Map visualizations
- Resources

Definitions

Definition of GIS

"a computer system that **analyzes** and **displays** geographically referenced information."
US Geological Survey

"a computer system for **capturing**, **storing**, **checking**, and **displaying** data related to positions on Earth's surface," enabling people to "more easily see, analyze, and understand patterns and relationships."
National Geographic - Education



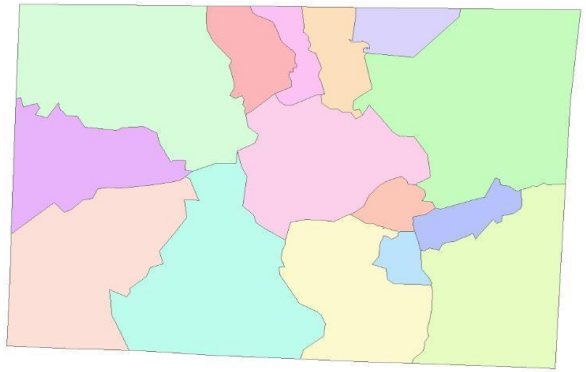
What is Geographic Information?

Place Names

Political Units: country, state, province
Natural Features: Mountain, river, lake

Administrative Units

ZIP code, Census tract, School district

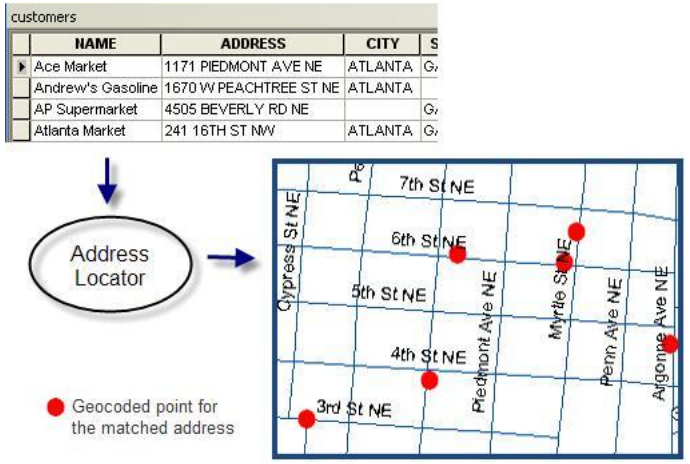


Street addresses

123 Scottybear Way
Highlander, CA 91954

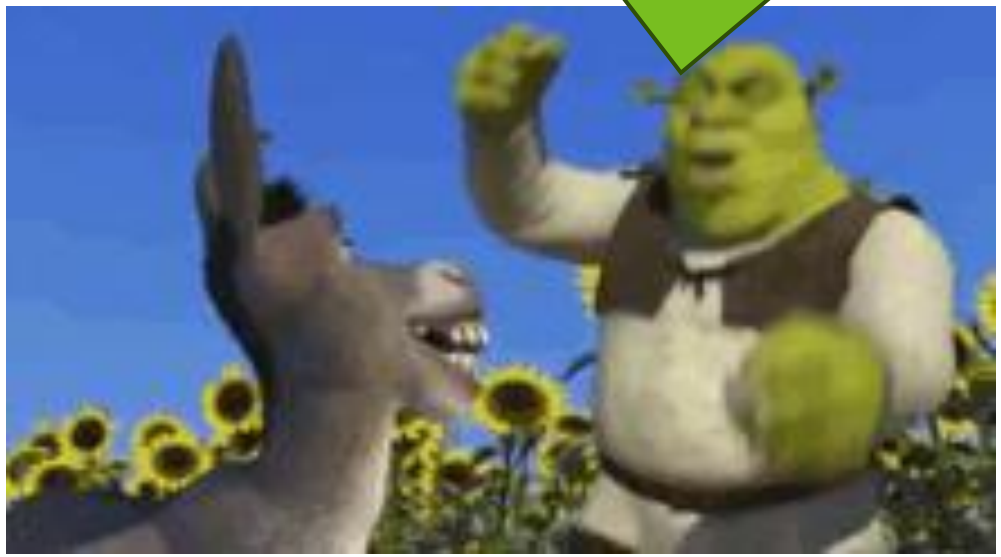
Coordinates

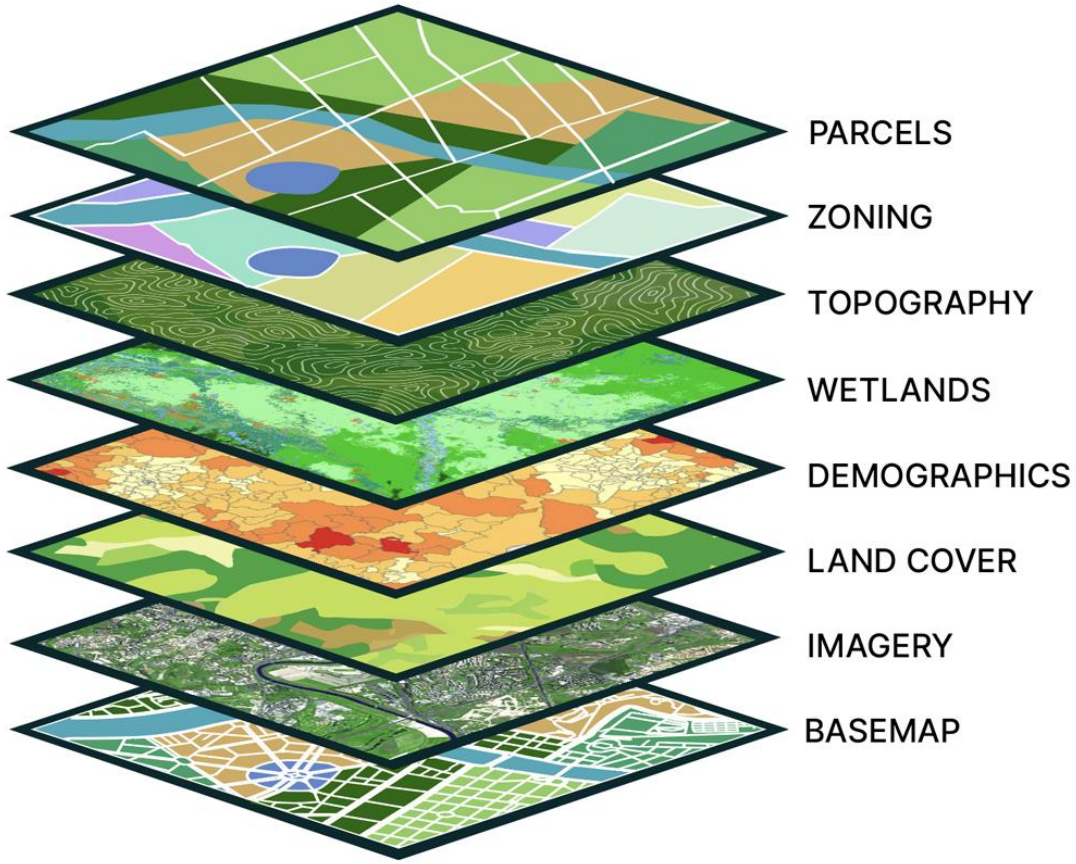
Latitude/Longitude
UTM Coordinates



Map Layers

GIS is like an onion, it has layers





https://www.earthdata.nasa.gov/learn/gis

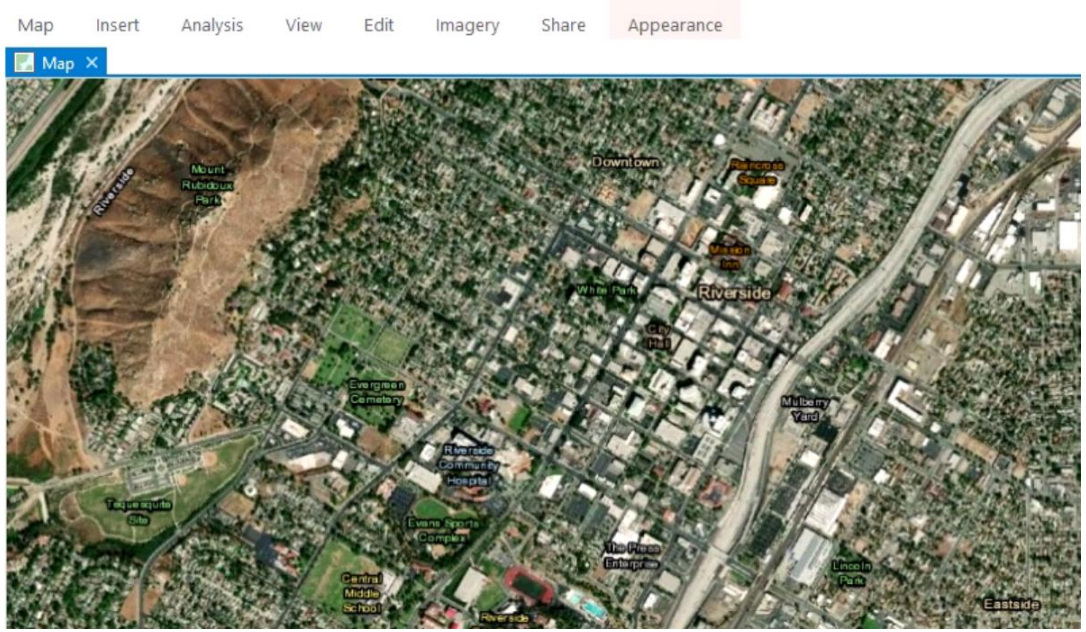
What is GIS used for?

A GIS can help answer questions about

- Location
  - Where are all the _____?
  - What are all the features in this place?
- Patterns and associations
  - How is _____ distributed across the area: randomly? In clumps? Gradually increasing/decreasing?
  - What else is present that might be affecting the pattern?
- Trends
  - How has the presence/distribution of _____ changed over time?

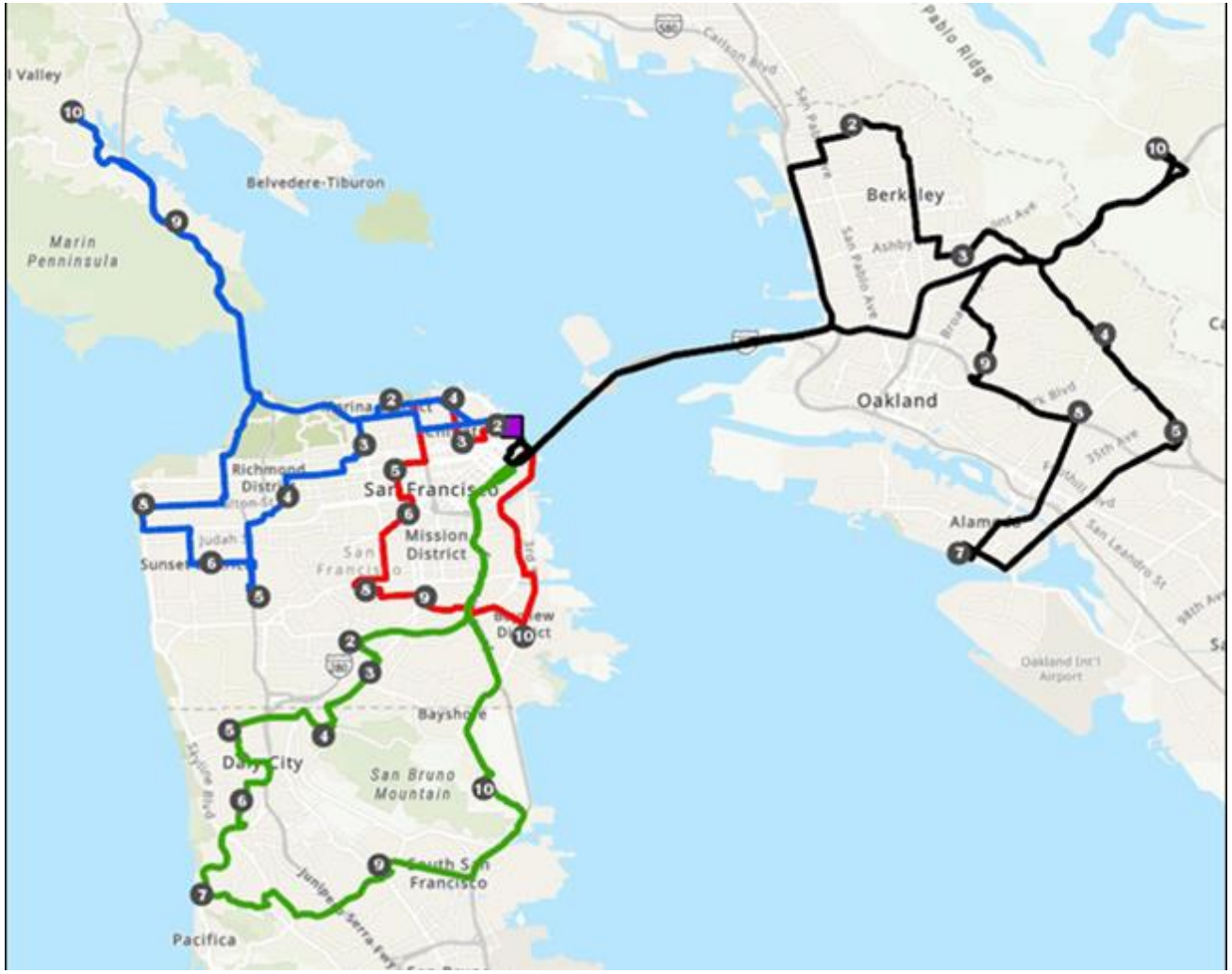
Map Views

Maps, Imagery



Associated Data

solutions.arcgis.com/local-government/help/homeless-point-time-counts/



Routing Analysis

Food delivery trucks to a chain of grocery stores

Deliver all required food

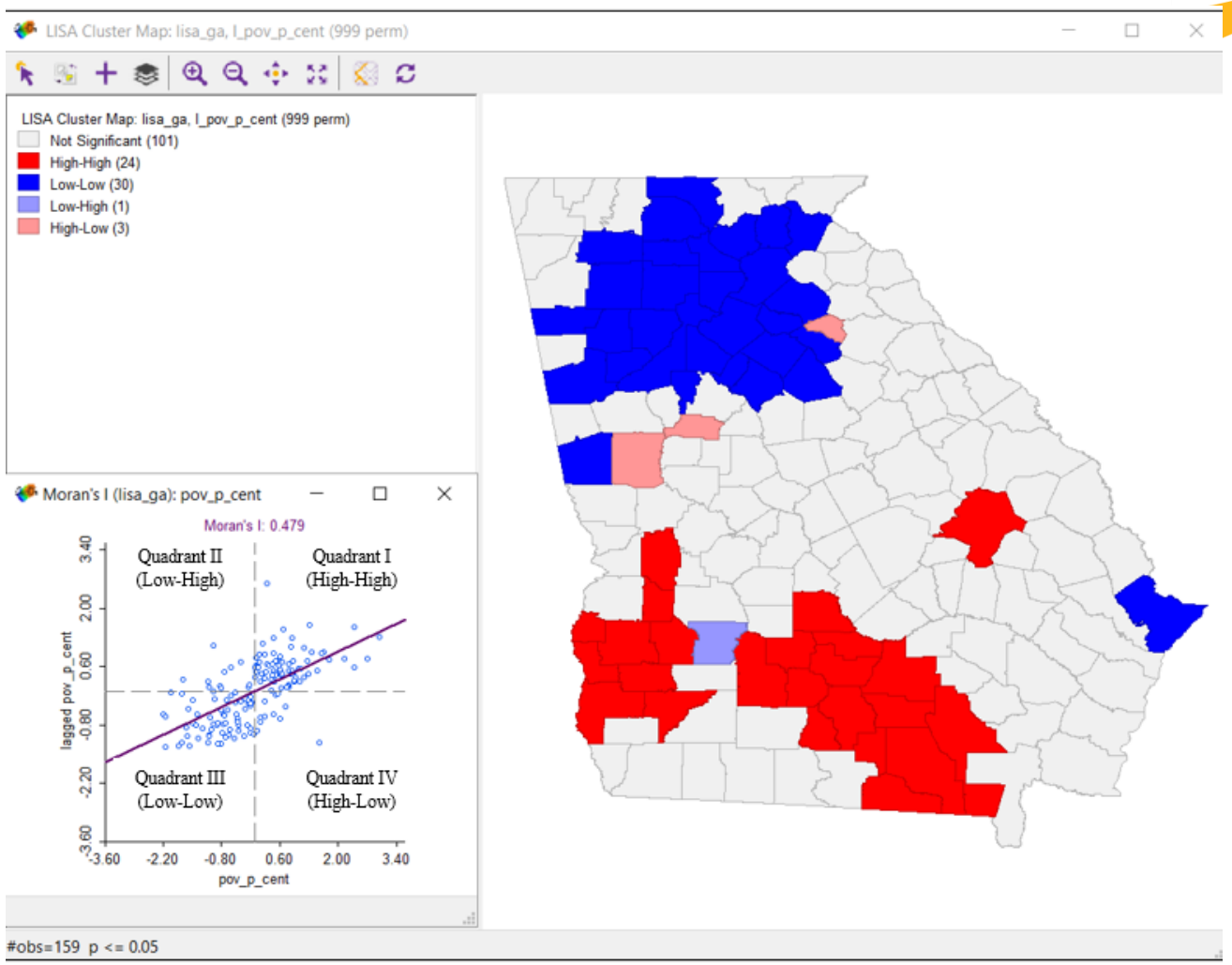
Minimize transportation costs

Adhere to vehicle capacities, lunch breaks, and maximum travel time

https://pro.arcgis.com/en/pro-app/latest/help/analysis/networks/what-is-network-analyst-.htm

Analyze Spatial Statistics

GIS can use spatial statistics to detect clusters and patterns, showing where similar values group together or where unusual outliers occur across space.

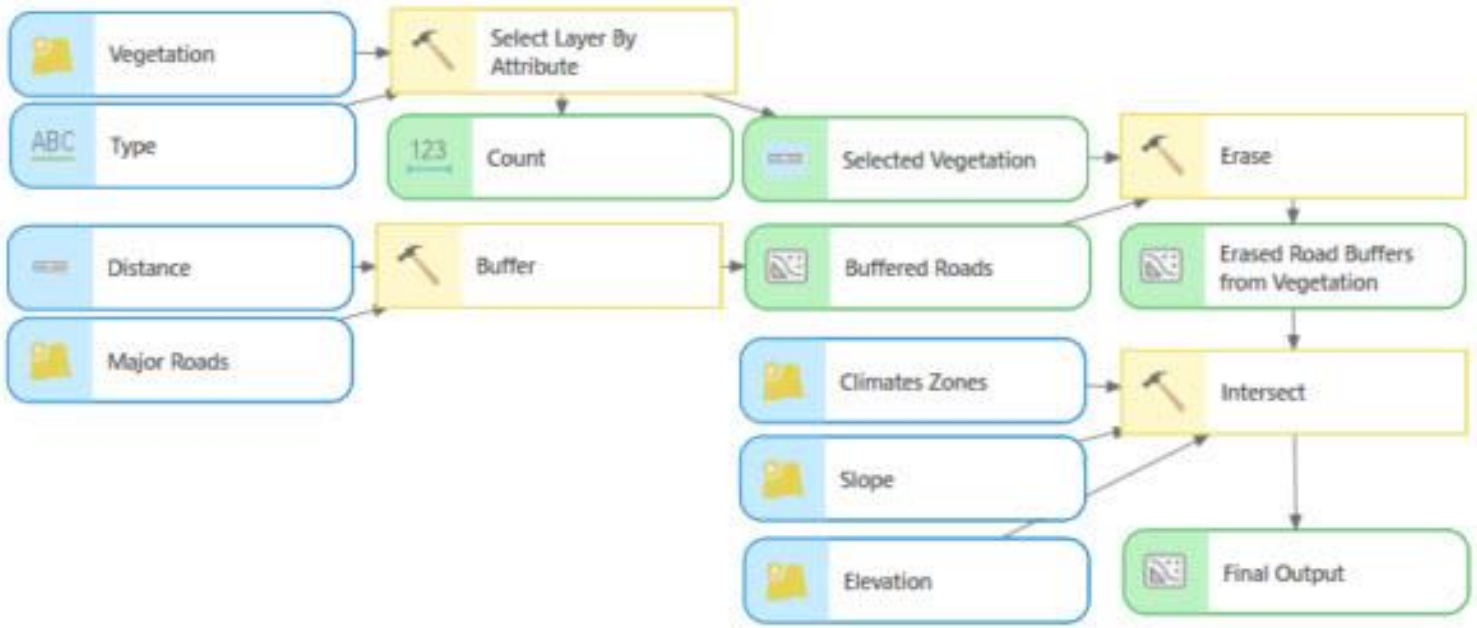


https://gistbok-ltb.ucgis.org/page/current/concept/AM-03-023

GIS Analysis Pipeline

Build a complex analysis pipeline that combines multiple operations

Potential habitat for an indigenous bird species



https://pro.arcgis.com/en/pro-app/latest/help/analysis/geoprocessing/modelbuilder/what-is-modelbuilder-.htm

Types of GIS Data

Vector

Raster

Vector Data

Longitude

Latitude

(x, y)

Point

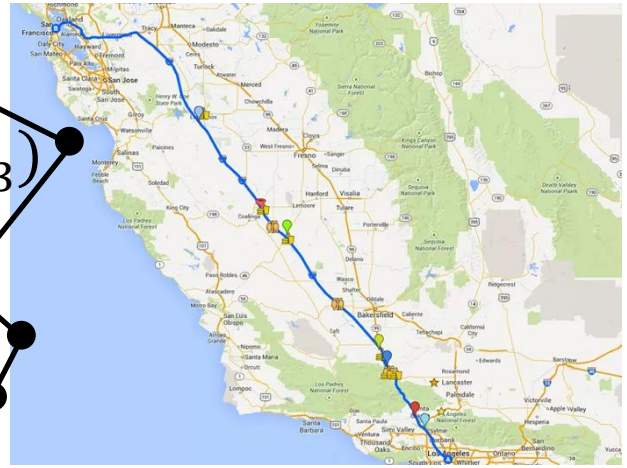
(x_1, y_1)

(x_2, y_2)

(x_3, y_3)

...

(x_n, y_n)



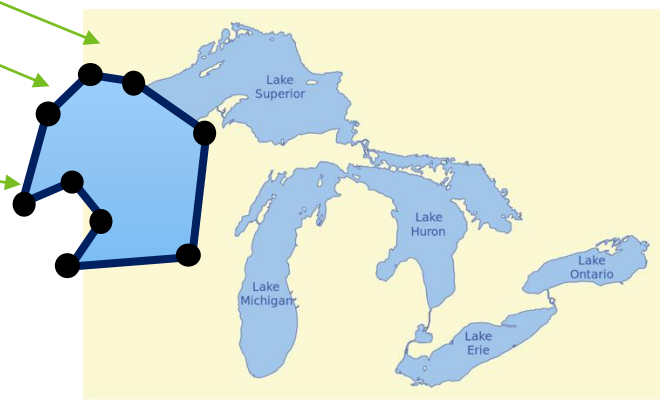
Line string or Polyline

$(x_1, y_1) = (x_n, y_n)$

(x_2, y_2)

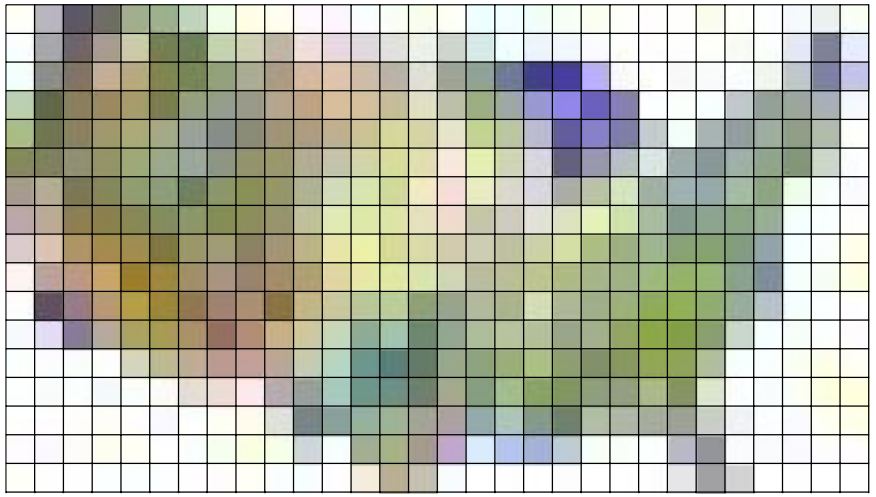
(x_3, y_3)

Polygon



GPS

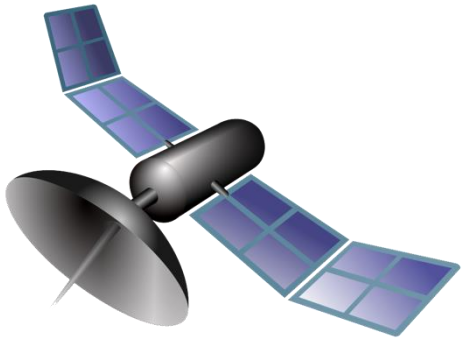
Raster Data



2D Array of values (pixels)

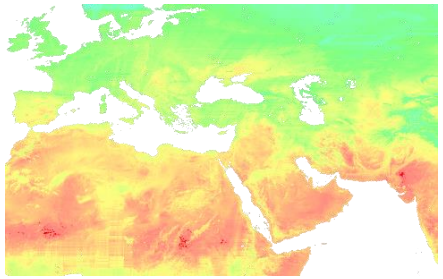
Drone

Camera



Satellite

Vegetation

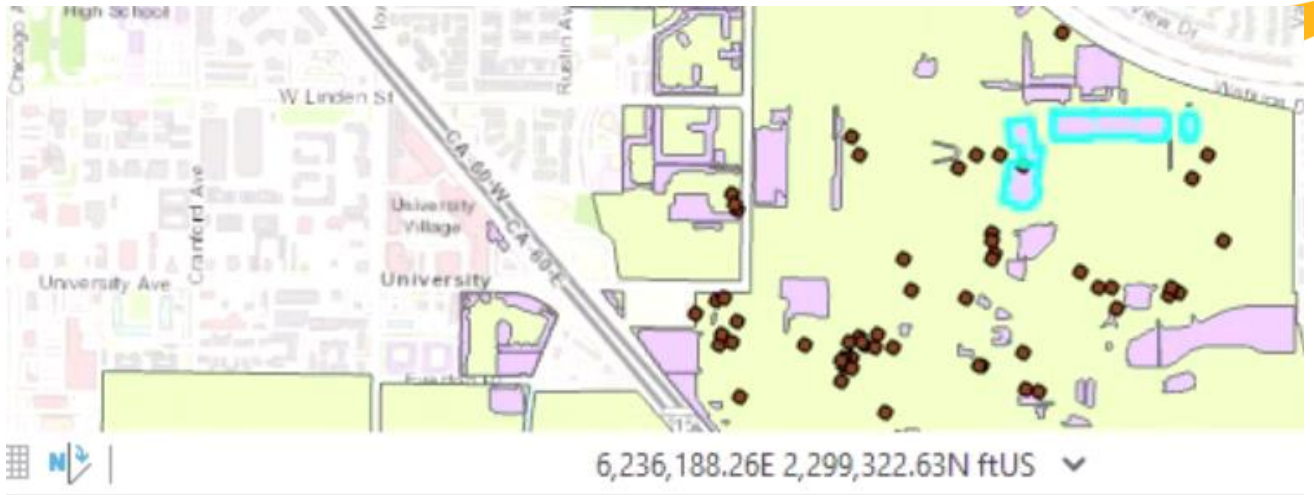


Temperature

Tabular Data

Records in the table are associated with features on the map by an ID number

You can display or analyze data in any of the columns (“fields”)



Selection: Zoom To Switch Clear Delete Copy

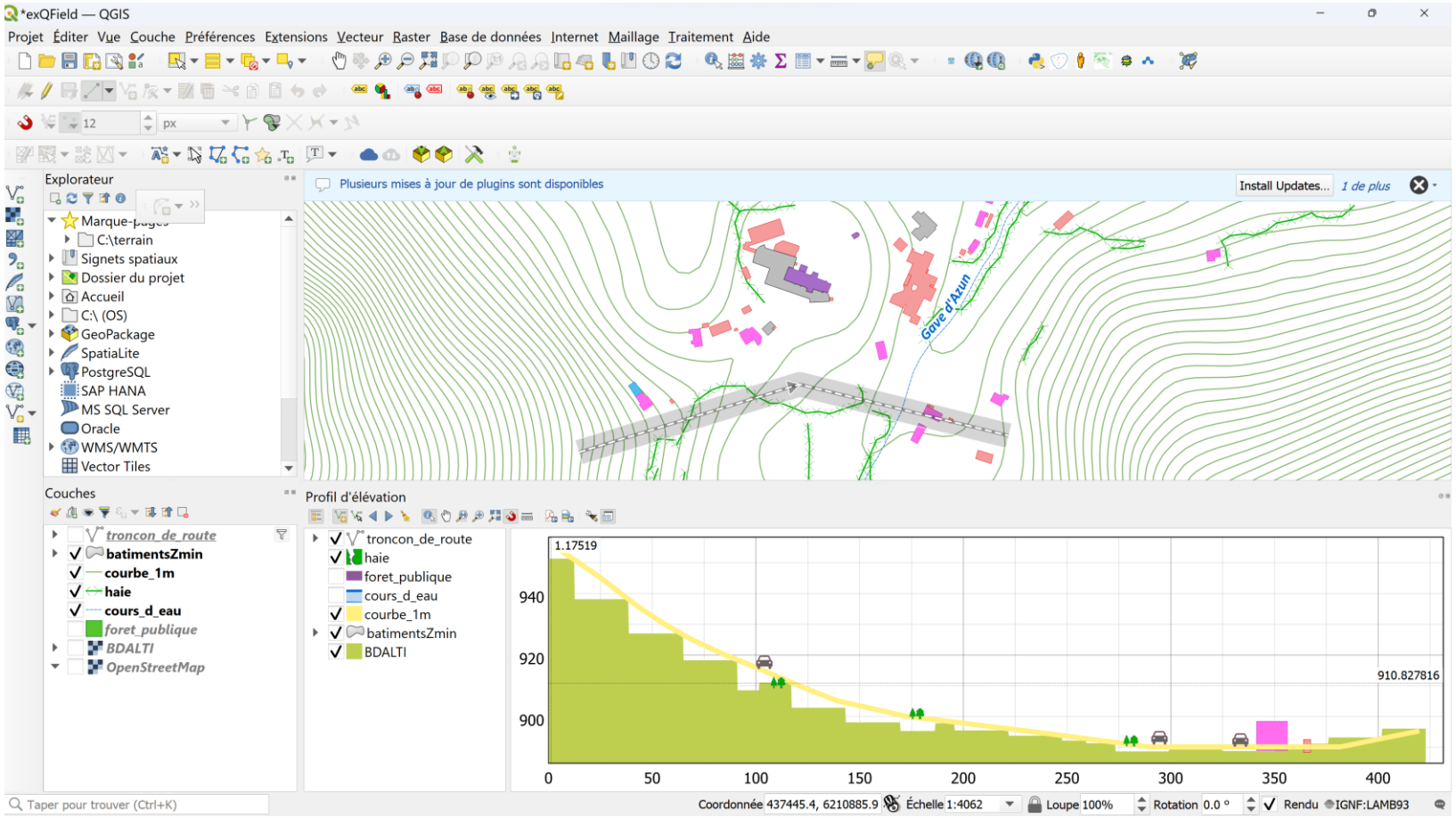
D	Lot_ID	Spaces	Lot_Type	Permit_Typ	Note
4	Lot9	137		Blue	
5	Lot22	324		Housing	
6	Lot21	427		Housing	
7	Lot7	42		Red	
8	Lot3	30		Red	
9	Lot8	55		Blue	

GIS Software Libraries

Types of GIS software

- Desktop applications
  - E.g., QGIS, ArcGIS, GRASS
- Web-based applications
  - E.g., ArcGIS Online, Carto, Mapbox
- Command line tools
  - E.g., GDAL, Geotools
- Software libraries
  - E.g., Geopandas, Geotools, Maplibre, Starlet

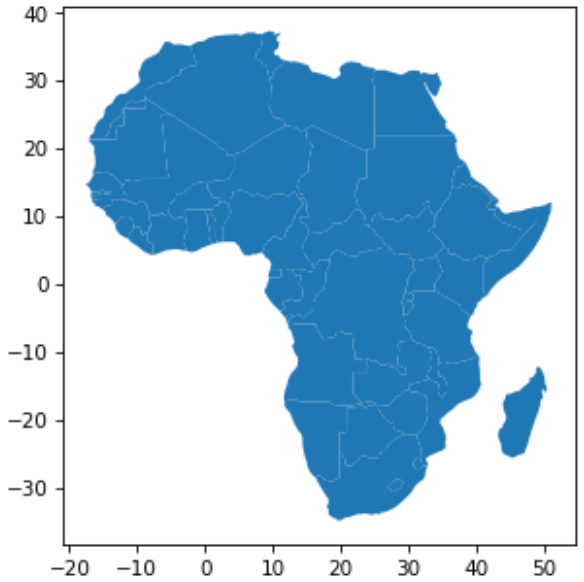
QGIS



https://www.sigterritoires.fr/index.php/en/how-to-create-a-topographic-profile-with-qgis-3-26-part-5-2/

Python APIs: GeoPandas

```
import geopandas as gpd
import matplotlib.pyplot as plt
world = gpd.read_file("ne_110m_admin_0_countries.zip")
# Filter: select countries in Africa
africa = world[world["CONTINENT"] == "Africa"]
# Plot the map
africa.plot()
plt.savefig("africa_map.png", dpi=75, bbox_inches="tight")
plt.close()
```



https://www.naturalearthdata.com/downloads/110m-cultural-vectors/

Command Line Tools

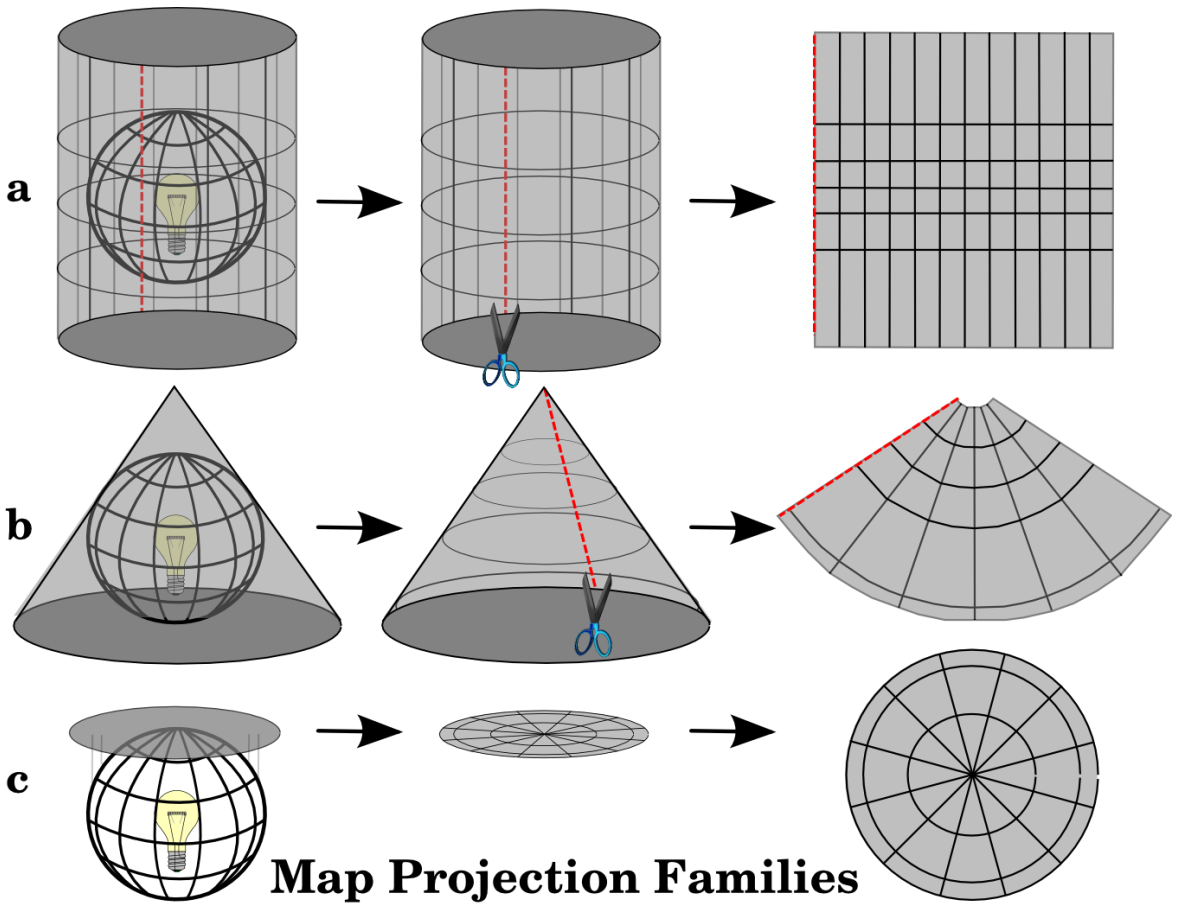
```
# Filter and save as a new file
ogr2ogr africa.geojson ne_110m_admin_0_countries.shp \
  -where "CONTINENT = 'Africa'"

# Plot to an image
gdal_rasterize -burn 0 -burn 156 -burn 222 \
  africa.geojson africa.tif

# Convert to PNG
gdal_translate africa.tif africa_map.png -of PNG
```

https://www.naturalearthdata.com/downloads/110m-cultural-vectors/

Projections and Coordinate Systems

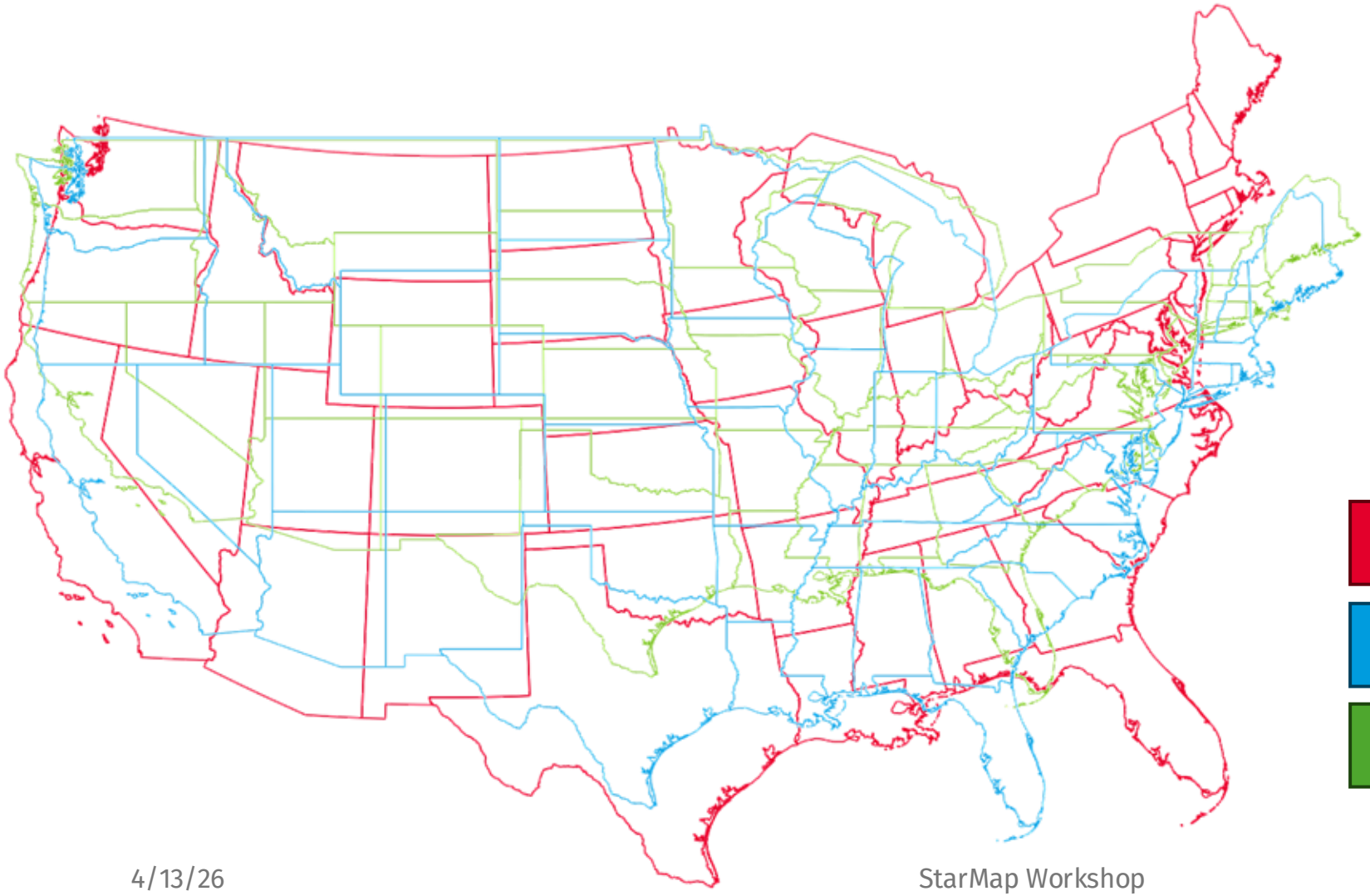
Map Projections

- GIS uses map projections to transform the curved Earth into a flat map
- Different projection surfaces (cylindrical, conic, planar) create different map shapes
- Every projection introduces some distortion (area, shape, distance, or direction)
- The choice of projection depends on the purpose of the map
- GIS allows users to select and switch projections for accurate analysis



https://docs.qgis.org/3.44/en/docs/gentle_gis_introduction/coordinate_reference_systems.html

Examples of Map Projections



Lambert Conformal Conic

Web Mercator

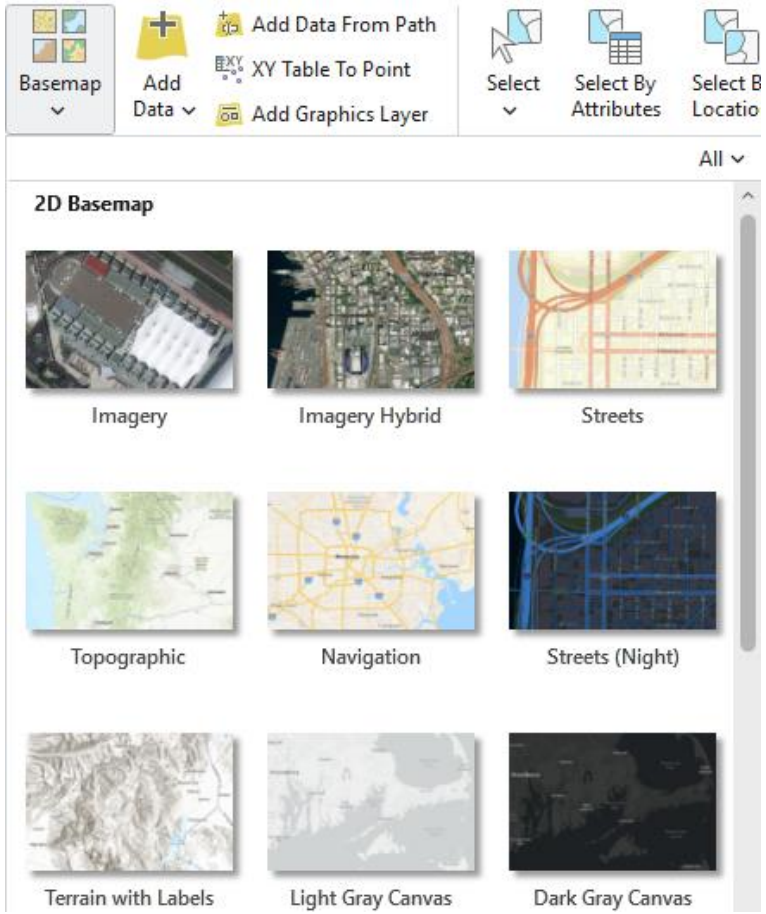
Un-projected (Lat, Lon) (WGS84)

Coordinate systems and implications

- Coordinate Systems - hundreds
  - Global / Spherical (latitude, longitude – angles)
  - Projected (e.g. UTM, State Plane, … - grid coordinates)
- If your data sources differ in projection or coordinate system, your data may not line up.
- GIS platforms can **reproject** on the fly, but the accuracy of analysis results will be in doubt.

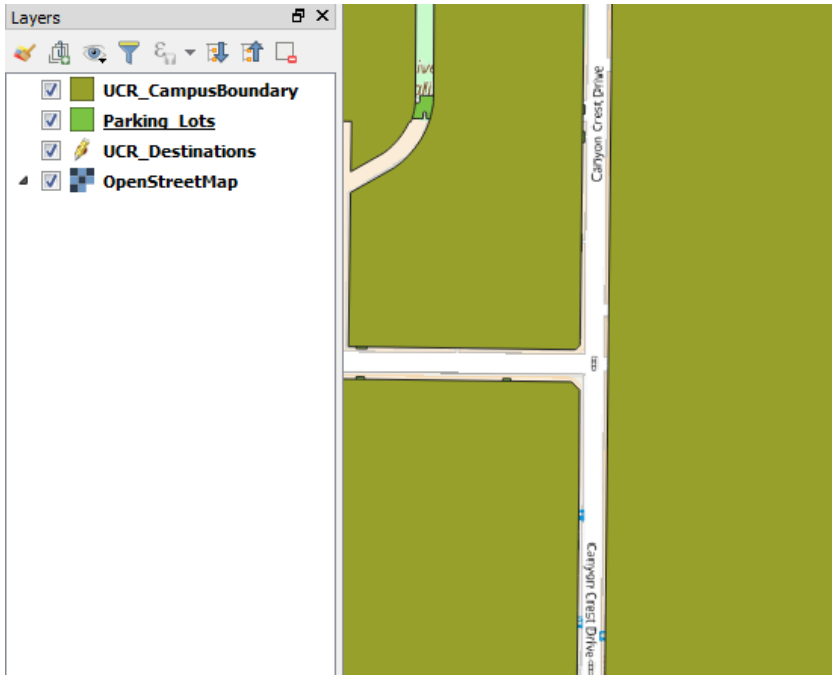
Map Visualizations

Choose a basemap

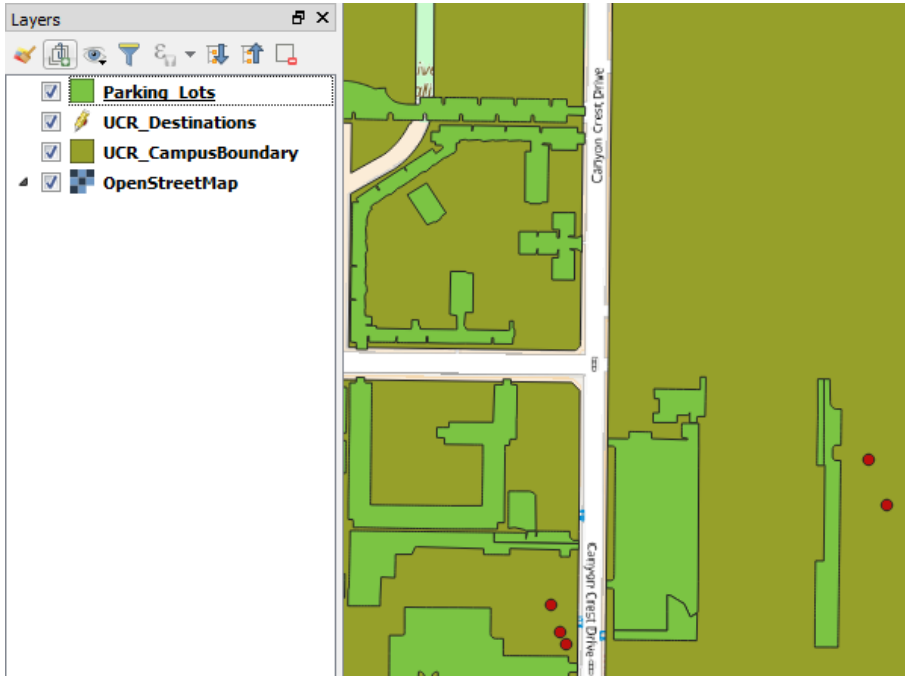


Add data; arrange the layers

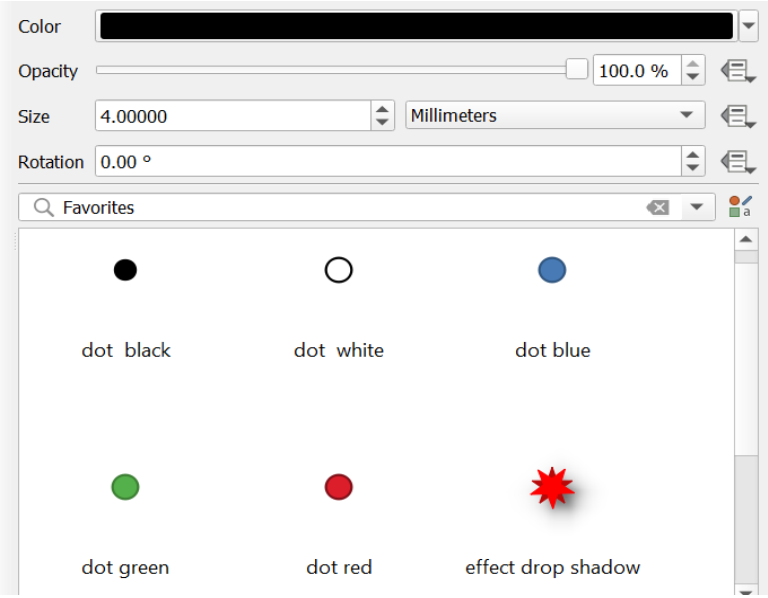
UCR_CampusBoundary at top

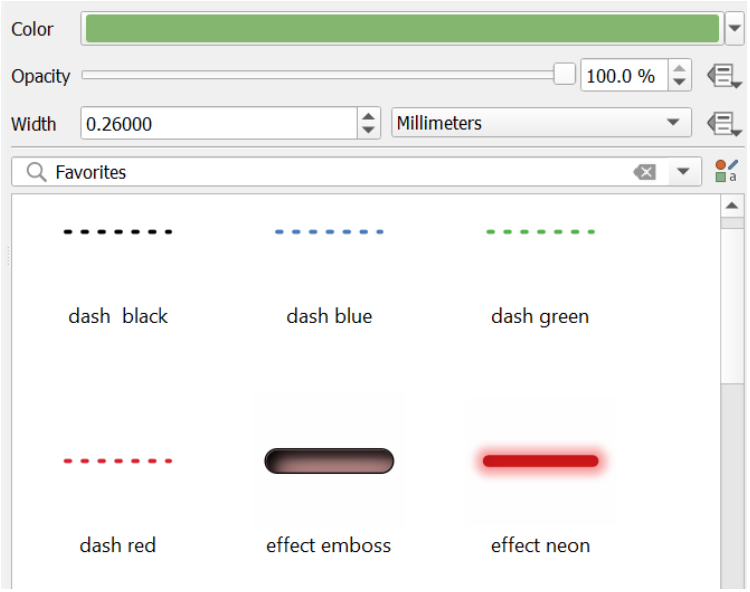


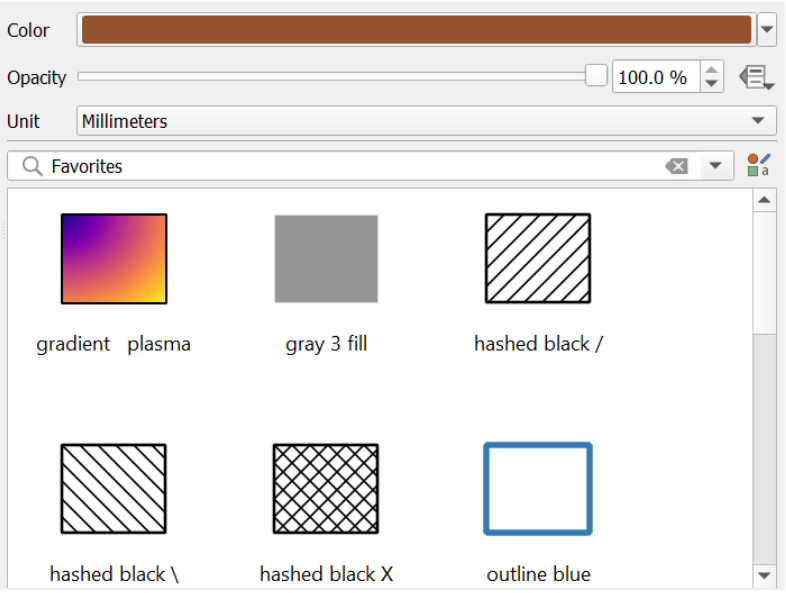
UCR_CampusBoundary in 3rd place



Change Symbology and Colors



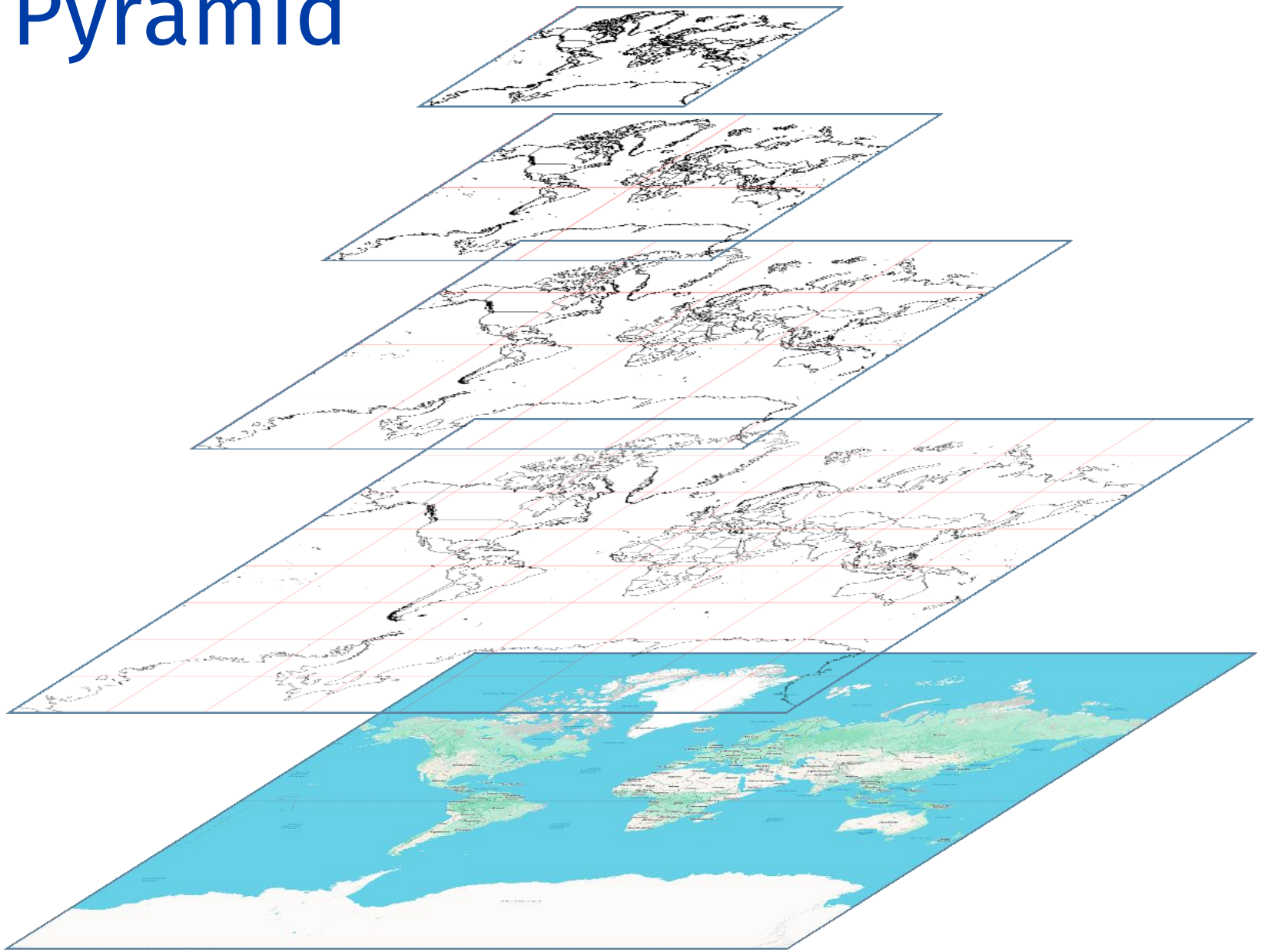




Web-based Maps and Big Data

- Data layers can be extremely big
- Web-based maps load partial data based on the current view
- At lower zoom levels:
  - Use aggregated and summarized view
- As users zoom in:
  - Load a more detailed version of the current view
- When uses pan around
  - Load a different region at the same zoom level
- The challenge is to generate and store billions of tiles

Tile Pyramid



GIS Data Sources

Where to find GIS Data?

- Data.gov
- UCR-Star: https://star.cs.ucr.edu
- County and city GIS portals
  - Riverside County: https://data.countyofriverside.us/browse
  - LA County: https://planning.lacounty.gov/maps-and-gis/gis-data/
  - Others
- Esri Living Atlas: https://livingatlas.arcgis.com/en/home/
- Big 10 Geoportal: https://geo.btaa.org

Resources

Further Readings

- Map school: https://mapschool.io/
- Introduction to GIS: https://volaya.github.io/gis-book/en/index.html
- Intro to GIS and Spatial Analysis: https://mgimond.github.io/Spatial/index.html
- Introduction to Geographic Information System: https://bookdown.org/tep/gis/introduction-to-gis.html
- Tutorials from Esri: https://learn.arcgis.com/en/paths/modern-gis-introduction-to-gis/
- Mapping, Society, and Technology: https://open.umn.edu/opentextbooks/textbooks/mapping-society-and-technology

Thank You!

Questions?